

Resolving A Vector

SYNOPSIS

Head-to-tail addition joins two vectors into one. *Resolving* a vector is that process in reverse: instead of combining two vectors into a resultant, we split *one* vector into two perpendicular parts — a *horizontal component* and a *vertical component* — that, added head-to-tail, reconstruct the original vector exactly. A vector that already lies flat along an axis needs no resolving at all: it's entirely one component and zero the other. Resolving is the tool that lets us add vectors algebraically instead of just diagrammatically — add the horizontal parts together, add the vertical parts together, and reassemble.

PERFORMANCE EXPECTATIONS By the end of this note, you will be able to...

- PE1 Recognize resolving as the reverse of head-to-tail addition** — splitting one vector into two perpendicular parts instead of combining two into one.
- PE2 Resolve a vector into its horizontal (x) and vertical (y) components** by drawing a right triangle: the component legs run along the axes, and the original vector is the hypotenuse.
- PE3 Recognize when a vector needs no resolving** — a vector already lying along the x-axis or y-axis is entirely one component; its other component is zero.
- PE4 Resolve two or more vectors, then add their components separately** — sum all the x-components together, and all the y-components together.
- PE5 State the component-addition rule** — $C_x = A_x + B_x$ and $C_y = A_y + B_y$ — and explain why adding components independently gives the same resultant as head-to-tail addition.



You already know this

Splitting one diagonal thing into two straight parts isn't new — you've done it before:



TV screen size

A "55-inch" TV is measured along the diagonal, but the screen really has a width and a height. The diagonal is one number standing in for two perpendicular ones.



Cutting across a field

Walk diagonally across a rectangular field instead of around the two edges — you still end up some distance east and some distance north, just covered both at once instead of one leg at a time.



Pushing a joystick diagonally

Push a game controller stick diagonally and the game doesn't see "diagonal" — it resolves your push into an x-push and a y-push and moves your character by both, separately.

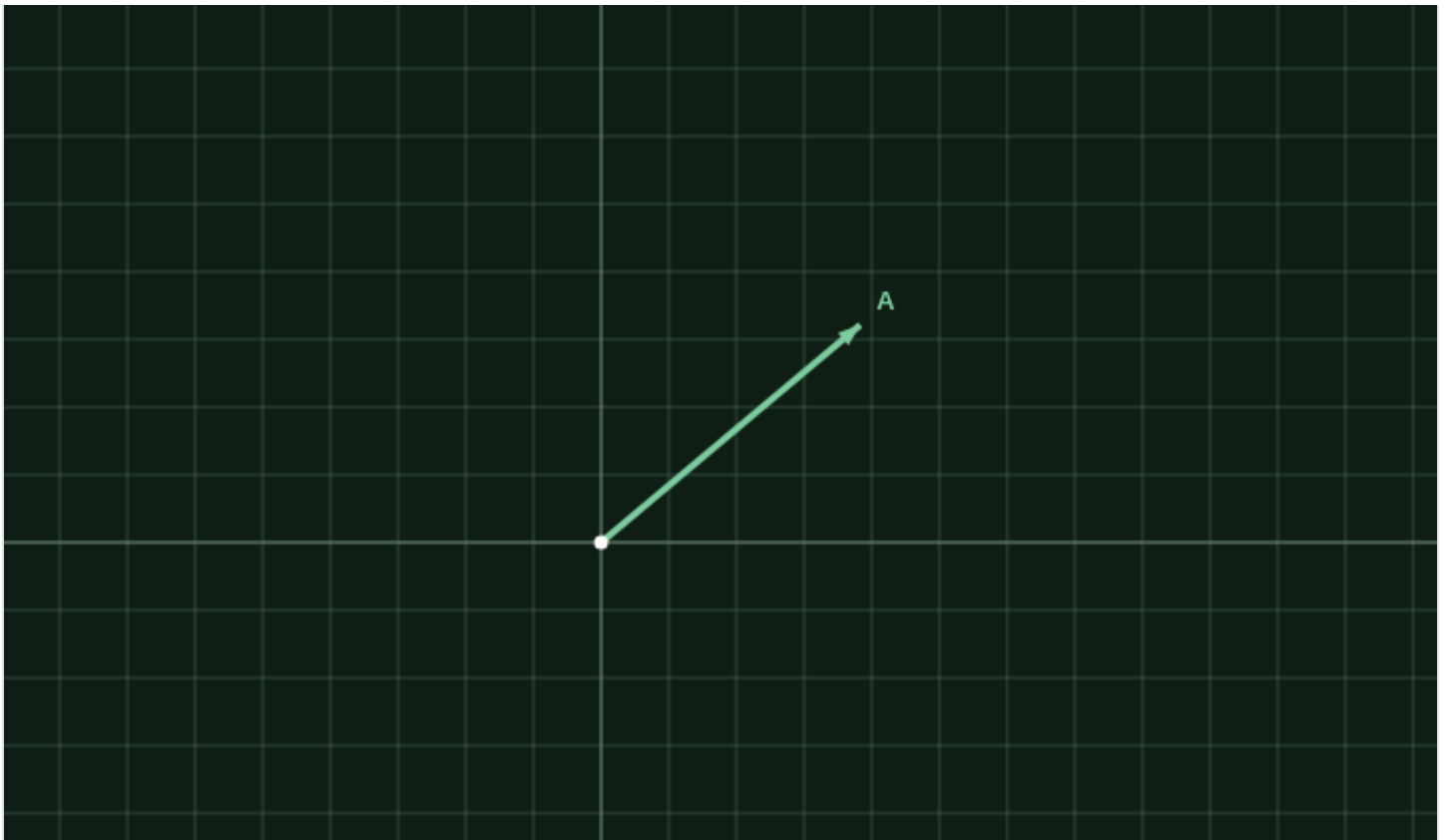
In this note, we work through

- I. What does it mean to "resolve" a vector, and how is that head-to-tail addition in reverse?
- II. How do we find a vector's horizontal and vertical components?
- III. What happens when we add two vectors' components separately — do we get the same resultant as adding the vectors head-to-tail?
- IV. Why does a vector already lying along an axis need no resolving?

Play with it: resolve a vector into components

Set vector **A**'s length and angle. **A_x** is how far it reaches horizontally; **A_y** is how far it reaches vertically. Together they form a right triangle with **A** itself as the hypotenuse. Try dragging the angle to exactly 0° or 90° — watch one component disappear.

► INTERACTIVE · VECTOR COMPONENT RESOLVER



Vector A — length

5.0 m

Vector A — angle

40°



Show components

↺ Reset

$A_x \approx -$

$A_y \approx -$

Predict A_x and A_y in your head first, then press "Show components" to check.

Predict first: a vector 5 m long points straight up, along the +y axis (90°). What is its horizontal (x) component?

5 m

0 m

2.5 m

It depends on the y-component

Watch: working through the note

WALKTHROUGH VIDEO • ~10 MIN

Video embed goes here — print the guided notes below before you press play.

Exit ticket

Four quick questions — check yourself before you close this note.

Q1. Resolving a vector into components means...

- ☐ Rotating it 90° .
- ☐ Splitting it into a horizontal part and a vertical part that add back to the original vector.
- ☐ Doubling its length.
- ☐ Combining it with another vector.

Q2. A vector lies exactly along the $+x$ axis. Its vertical (y) component is...

- ☐ Equal to its magnitude.

- ☐ Zero.
- ☐ Impossible to determine.
- ☐ Always negative.

Q3. Vector A has $A_x = 6$ m and vector B has $B_x = -2$ m. If $C = A + B$, what is C_x ?

m

Q4. Adding vectors by components — $C_x = A_x + B_x$ and $C_y = A_y + B_y$ — works because...

- ☐ The horizontal and vertical directions are independent of each other, so components along each axis add like ordinary numbers.
- ☐ It's a shortcut that only works for right angles between A and B.
- ☐ It only works when A and B point the same way.
- ☐ Vectors don't actually add — only their components do.

Check my answers

Clear

Your notes

READ ANYTIME

This note (PDF)

A print-friendly PDF of this page, for offline reading or a quick reference while you work the guided notes.

AFTER THE MODULE

Completed notes (PDF)

Released when Module 2 closes — check your work against it.

